

12/08/26

Signal Processing

- Recap: DFT, DTFT, Z-transform ✓
- Revisit discrete-time signals ✓
 - $\delta[n]$, $u[n]$
 - $A\alpha^n$ (real and complex)
 - Sinusoids: $A \cos(\omega_0 n + \phi)$; indistinguishability
 - Periodic signals
- Revisit discrete-time systems
 - Memoryless, linear, time-invariant, causal, stable
 - Linear time-invariant

- Revisit discrete time signals:

$$\delta[n] = \begin{cases} 1, & n=0, \\ 0, & n \neq 0 \end{cases} \quad n \in \mathbb{Z}$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot \delta[n-k] \quad (\text{arbitrary discrete-time signal})$$

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases} \quad \bullet \quad u[n] = \sum_{k=-\infty}^n \delta[k]$$

$$x[n] = A \cdot \alpha^n$$

- Real case: $A, \alpha \in \mathbb{R}$. The signal either increases, decreases or stays constant depending on our choice of α .
 $(\alpha > 1)$ $(\alpha < 1)$
 $(\alpha = 1)$

- Complex case: $A, \alpha \in \mathbb{C}$

$$\text{Let } A = |A| e^{j\phi}$$

$$\alpha = |\alpha| e^{j\omega_0}$$

$$x[n] = |A| \cdot |\alpha|^n \cdot e^{j\phi} \cdot e^{j\omega_0 n} = |A| \cdot |\alpha|^n \cdot e^{j(\omega_0 n + \phi)}$$

- "Indistinguishability" of certain frequencies in the discrete-time complex

exponential.

- If $\omega_1 = \omega_0 + 2\pi$, $\omega_1 > \omega_0$. However, ω_0 and ω_1 are indistinguishable from one another.

$$\begin{aligned} \text{i.e.; } |A||\alpha|^n e^{j(\omega_0 n + \phi)} &= |A||\alpha|^n e^{j(\omega_1 n + \phi)} \\ &= |A||\alpha|^n e^{j(\omega_0 n + 2\pi n + \phi)} \\ &= |A||\alpha|^n e^{j(\omega_0 n + \phi)} \cdot \cancel{e^{j2\pi n}}^1 \end{aligned}$$

- Periodic signals: A signal $x[n]$ is said to be periodic if $\exists N$ (positive integer) such that $x[n+N] = x[n]$ for all $n \in \mathbb{Z}$.

a) Is $\cos(\pi n)$ periodic? $\rightarrow$ find an N for which $\cos(\pi n + N) = \cos(\pi n)$
 $\rightarrow N = 2$.

b) Is $\cos(n)$ periodic? $\cos(n + 2\pi k) = \cos(n)$. However $\nexists$ integer $N = 2\pi k$.

c) Is $\cos(\omega_0 n)$ periodic? If $\cos(\omega_0(n+N)) = \cos(\omega_0 n)$, then

$$\omega_0 N = 2\pi k. \quad \text{i.e. } \omega_0 = \frac{2\pi k}{N}$$

By implication, (c) extends to $A\alpha^n$ where $A, \alpha \in \mathbb{C}$, $|\alpha| = 1$.

- Revisit discrete-time systems: A discrete-time system is a mapping from a signal $x[n]$ to a desired output signal $y[n]$

$$y[n] = \mathcal{T}\{x[n]\}$$